

Day 3 JavaScript Assignment

Instructions

- Write clean, well-commented JavaScript code.
- Display output using `console.log()` or `document.write()`.
- Validate inputs wherever appropriate.

Problem 1: Employee Salary Calculator

A startup wants an internal salary estimator.

Inputs:

- Employee Name
- Base Salary
- Monthly Bonus
- Tax Percentage

Requirements:

1. Calculate Gross Salary = Base Salary + Bonus.
2. Calculate Tax Amount.
3. Calculate Net Salary.
4. If Net Salary ≥ 100000 display "High Income Employee".
5. Else if Net Salary ≥ 50000 display "Mid Income Employee".
6. Otherwise display "Entry Level Employee".

Expected Output:

Employee Name, Gross Salary, Tax Amount, Net Salary, Category.

Problem 2: Online Shopping Checkout System

Create the billing logic for an e-commerce website.

Inputs:

- Customer Name
- Product Price
- Quantity
- Membership (Yes/No)

Requirements:

1. Calculate Total Amount.
2. Members receive 10% discount.
3. If final amount ≥ 5000 provide Free Shipping otherwise Shipping = ₹200.
4. Print Final Payable Amount.
5. Display whether customer qualifies for Premium Purchase ($\geq ₹10000$).

Problem 3: Student Scholarship Eligibility

A university wants to automate scholarship screening.

Inputs:

- Student Name
- Percentage
- Annual Family Income

Requirements:

1. Percentage ≥ 90 and Income < 300000 $\Rightarrow$ 100% Scholarship.
2. Percentage ≥ 80 and Income < 500000 $\Rightarrow$ 50% Scholarship.
3. Percentage $\geq 70 \Rightarrow$ 25% Scholarship.
4. Otherwise Not Eligible.
5. Print final decision with student details.

Problem 4: Electricity Bill Generator

Develop a billing system for a power company.

Inputs:

- Consumer Name
- Units Consumed

Requirements:

1. First 100 units = ₹5/unit
2. Next 200 units = ₹7/unit
3. Remaining units = ₹10/unit
4. If total bill $> ₹5000$ give 5% rebate.
5. Print bill amount and bill status (Low/Medium/High based on amount).

Problem 5: Loan Eligibility Checker

Create a loan pre-screening system for a bank.

Inputs:

- Applicant Name
- Age
- Monthly Salary
- Existing EMI

Requirements:

1. Calculate EMI Ratio = Existing EMI / Monthly Salary $\times 100$.
2. Age must be between 21 and 60.
3. Salary must be at least ₹30000.
4. EMI Ratio must not exceed 40%.
5. Display:
 - Approved
 - Conditionally Approved
 - Rejected

Also print the reason for rejection if any.